# Explainable Deep Learning-Based Skin Condition Classification and Severity Estimation Using Transfer Learning and Grad-CAM Visualization

by

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
M.Sc. in Computer Science

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# Declaration

It is hereby declared that

1. The thesis submitted is my own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. I have acknowledged all main sources of help.

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# Approval

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# Ethics Statement

This study focuses on the development and evaluation of deep learning models for skin condition classification and severity estimation using image datasets. The research relies on image data for computational analysis and does not involve direct clinical examinations, medical interventions, or experiments on human participants conducted by the researcher. The datasets are used for academic and research purposes, subject to their respective access conditions and terms of use. Appropriate care is taken to acknowledge the original data sources and cite relevant publications.

The study aims to support the interpretation of model predictions through Gradient-weighted Class Activation Mapping (Grad-CAM). However, the generated predictions and visual explanations are intended for research purposes only and should not be considered a substitute for professional medical diagnosis or clinical judgment. The limitations of the models and the possibility of incorrect predictions are acknowledged. The research does not provide treatment recommendations or claim clinical validity.

All reported results are presented to reflect the models' observed performance on the selected datasets. The study does not intentionally manipulate evaluation results or misrepresent model capabilities. Any future work involving patient recruitment, identifiable medical images, clinical validation, or deployment in a healthcare setting will require appropriate ethical review, permissions, and safeguards before implementation.

# Abstract

Accurate identification of skin conditions is important for understanding skin health and supporting appropriate care. Deep learning techniques can classify skin conditions from images, while transfer learning enables the use of pre-trained models for this task. However, model performance may vary depending on the network architecture and the division of data into training, validation, and testing sets. Furthermore, visual explanations can help identify image regions that influence model predictions. This study evaluates EfficientNetB0, MobileNetV2, and ResNet50 for image-based skin condition classification and severity identification, using Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize regions associated with their predictions. The developed system classifies images into 12 labels across four categories: Acne, Burn, Eczema, and Normal skin. The first three categories include mild, moderate, and severe levels, whereas the Normal category comprises dry, normal, and oily skin types. The models are evaluated using different dataset splitting ratios and compared based on classification performance and inference time. Among the evaluated configurations, ResNet50 with a 90% training, 5% validation, and 5% testing split achieved the highest recorded accuracy of 86.84%, precision of 87.22%, recall of 86.84%, F1-score of 86.79%, and ROC-AUC of 99.27%, with an inference time of 72.42 ms per image. Grad-CAM provides visual explanations by highlighting image regions associated with model predictions. These findings demonstrate the potential of transfer learning and explainable deep learning for skin condition classification and severity assessment, while highlighting the need for evaluation on more diverse and independent datasets to assess generalization to unseen images.

**Keywords:** Skin condition classification; Transfer learning; Deep learning; Explainable artificial intelligence (XAI); Grad-CAM; Convolutional neural networks

# Dedication

I dedicate this thesis to my beloved parents and family, whose unconditional love, countless sacrifices, and unwavering support have been the foundation of my academic journey. Their encouragement, patience, and belief in me have given me the strength to overcome challenges and continue striving toward my goals. This achievement would not have been possible without their continuous support and inspiration.

I would also like to dedicate this work to my thesis supervisor, **Mr. Annajiat Alim Rasel**, for his guidance, encouragement, and valuable feedback throughout my research journey. His support has been an important part of my academic development and has motivated me to improve my work.

Finally, I dedicate this thesis to everyone who has inspired me to pursue knowledge, remain persistent in the face of difficulties, and continue learning and growing throughout my academic journey.

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# Chapter 1

# Introduction

## 1.1 Background

Skin conditions affect people of different ages and can influence both physical health and quality of life. Conditions such as acne, eczema, and burns may vary in their appearance and severity, making accurate identification important for appropriate care. However, access to dermatological services can be difficult in some areas because of the limited availability of specialists, consultation costs, and geographical barriers. These challenges have encouraged researchers to investigate computer-aided systems that can analyze skin images and assist with the identification of skin conditions. Such systems may provide useful preliminary information, although they cannot replace professional medical diagnosis.

Recent developments in artificial intelligence have created new opportunities for analyzing medical images. Convolutional Neural Networks (CNNs) can learn visual features from images and use them to distinguish between different categories. Transfer learning provides another approach by allowing models trained on large image datasets to be adapted to specific classification tasks. Models such as EfficientNetB0, MobileNetV2, and ResNet50 offer different architectural approaches to feature extraction and computational efficiency. However, their performance may vary depending on the dataset, the classification task, and the proportions of data used for training, validation, and testing. Therefore, comparing these models under different data-splitting configurations can help identify a suitable approach for skin condition and severity classification.

Although deep learning models can classify images, their predictions do not always make it clear which visual features influenced the results. This limitation is particularly important in medical image analysis, where understanding a model's predictions is necessary for evaluating its usefulness. Explainable Artificial Intelligence (XAI) methods can help address this issue by providing visual information about a model's decision-making process. One such method is Gradient-weighted Class Activation Mapping (Grad-CAM), which produces a heatmap highlighting image regions that contribute to a selected model prediction. In this study, Grad-CAM is used to examine the predictions made by the evaluated models, alongside a comparison of their classification performance and training time. By investigating different model architectures and dataset splits, this research aims to assess their suitability for skin condition classification and severity identification while providing visual explanations of the predictions.

## 1.2 Problem Statement

Conventional diagnosis of skin conditions primarily depends on visual examination by experienced dermatologists. Although this approach is effective, it may not always be readily available, especially in remote or resource-limited areas. Delays in diagnosis can result in disease progression and reduced treatment effectiveness. In addition, many existing automated skin analysis systems focus solely on disease classification without providing information about severity or explaining how predictions are made. Another important limitation is the lack of practical tools for monitoring changes in skin conditions over time. Patients undergoing treatment often rely on subjective visual observation to determine whether their condition is improving. An intelligent system capable of estimating severity levels from skin images can provide a consistent and objective method for comparing images captured at different stages of treatment. Such a system can assist healthcare professionals and patients in monitoring disease progression or recovery while improving confidence through visual explanations.

## 1.3 Research Objectives

The primary objective of this research is to develop an explainable deep learning framework for automatic skin condition classification and severity estimation using transfer learning techniques.

The specific objectives of this study are:

- To develop an automated skin condition classification system capable of identifying multiple skin conditions from image data.
- To estimate the severity or condition level associated with each identified skin condition.
- To implement and compare multiple transfer learning models, including EfficientNetB0, MobileNetV2, and ResNet50, for skin condition classification.
- To generate confidence scores for every skin condition–severity label, providing a comprehensive prediction rather than a single output.
- To incorporate Grad-CAM visualization to explain the model's decision-making process by highlighting influential image regions.

To provide a framework that can assist in monitoring changes in skin condition severity over time by comparing predictions obtained from images captured during different stages of treatment.

## 1.4 Scope of the Study

This research focuses on the classification of four skin condition categories: Acne, Burn, Eczema, and Normal skin. The study utilizes an image dataset with multiple severity or condition labels and employs transfer learning to train deep learning models. Three pre-trained CNN architectures - EfficientNetB0, MobileNetV2, and

ResNet50 - are comparatively evaluated to determine their effectiveness for the proposed task.
The developed framework predicts confidence scores for each skin condition - severity label and provides Grad-CAM visualizations to improve model interpretability. The proposed system is intended to function as a computer-aided decision support tool and is not designed to replace professional medical diagnosis. Instead, it aims to assist users by providing preliminary assessments and visual explanations while supporting the monitoring of changes in skin condition severity throughout the treatment process.

## 1.5 Organization of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 presents a review of the relevant literature on skin condition classification, transfer learning, and explainable artificial intelligence. Chapter 3 describes the research methodology, including dataset preparation, preprocessing, model development, training procedures, and evaluation metrics. Chapter 4 presents the experimental results, comparative analysis of the selected transfer learning models, and Grad-CAM visualizations. Finally, Chapter 5 concludes the study by summarizing the major findings, discussing limitations, and providing recommendations for future research.

# Chapter 2

# Literature Review

## 2.1 Introduction

Skin diseases are among the most common medical conditions worldwide and affect individuals of all ages. Early detection and continuous monitoring of skin conditions are essential for preventing complications and improving treatment outcomes. Recent advances in Artificial Intelligence (AI), particularly Deep Learning (DL), have significantly improved the automatic analysis of medical images. In dermatology, deep learning models have demonstrated remarkable performance in identifying skin diseases from digital images, often achieving results comparable to those of experienced clinicians.
This chapter presents the theoretical background relevant to the proposed research. It discusses skin conditions and severity estimation, convolutional neural networks, transfer learning, explainable artificial intelligence, and Grad-CAM visualization. Finally, related studies and the research gap are presented to establish the motivation for the proposed framework.

## 2.2 Skin Conditions and Severity Estimation

Skin diseases represent a diverse group of medical conditions that affect the skin's appearance, structure, and function. They may result from infections, inflammation, allergies, burns, environmental exposure, or genetic factors. Accurate identification of these conditions is important because delayed or incorrect diagnosis may lead to disease progression and ineffective treatment.

This research focuses on four skin condition categories:

- Acne
- Eczema
- Burn
- Normal Skin

Each condition exhibits distinct visual characteristics. Acne is commonly characterized by pimples, papules, pustules, and inflammation. Eczema often appears as

dry, itchy, inflamed patches with varying degrees of redness. Burn injuries present different levels of tissue damage depending on their severity, while normal skin includes healthy skin conditions with variations such as dry, normal, and oily skin. Beyond identifying the skin condition, estimating its severity provides additional clinical value. Severity assessment enables healthcare professionals and patients to monitor disease progression and evaluate treatment effectiveness over time. Instead of simply identifying the disease category, a severity-aware system offers more meaningful information for long-term patient management.

## 2.3 Deep Learning for Medical Image Classification

Deep learning has become one of the most successful approaches for medical image analysis. Unlike traditional machine learning techniques that rely on manually engineered features, deep learning models automatically learn hierarchical feature representations directly from raw image data.

Convolutional Neural Networks (CNNs) are particularly effective for image classification because they automatically learn low-level features such as edges and textures before extracting higher-level semantic information. Through multiple convolutional layers, pooling operations, and nonlinear activation functions, CNNs progressively identify complex visual patterns associated with different disease categories.

Medical image classification using deep learning has been successfully applied in various healthcare domains, including radiology, pathology, ophthalmology, and dermatology. In skin image analysis, CNN-based models have shown promising performance in distinguishing multiple skin diseases using dermoscopic and clinical photographs.

Despite these achievements, training deep neural networks from scratch generally requires very large annotated datasets and substantial computational resources. This limitation has encouraged researchers to adopt transfer learning techniques that leverage knowledge learned from large-scale image datasets.

## 2.4 Transfer Learning

Transfer learning is a deep learning technique in which knowledge acquired from one task is transferred to another related task. Instead of training a convolutional neural network from scratch, a model pre-trained on a large dataset such as ImageNet is adapted to solve a new classification problem.

Pre-trained models have already learned general visual features including edges, textures, shapes, and object structures. These learned representations can be reused for medical image classification, reducing training time and improving model performance, particularly when the available medical dataset is relatively small.

In this research, three transfer learning architectures were investigated:

- EfficientNetB0
- MobileNetV2

- ResNet50

Each model was initialized using ImageNet pre-trained weights, after which a custom classification head was added. During training, transfer learning enabled the models to adapt their learned visual representations to skin condition classification while reducing computational requirements.
Transfer learning has become one of the most widely adopted approaches in medical image analysis because it improves convergence speed, reduces overfitting, and achieves competitive performance even with limited annotated datasets.

## 2.5 Explainable Artificial Intelligence (XAI)

Although deep learning models often achieve high classification accuracy, they are frequently criticized for functioning as "black-box" systems because their decision-making processes are difficult to interpret. In medical applications, explainability is particularly important because clinicians require evidence supporting automated predictions before trusting AI-assisted diagnostic systems.
Explainable Artificial Intelligence (XAI) refers to a collection of techniques designed to make artificial intelligence models more transparent and understandable. XAI enables users to identify which image regions or features contributed most significantly to a model's prediction.

In healthcare, explainability provides several important benefits:

- Improves clinician confidence in AI-assisted diagnosis.
- Enhances transparency of automated decision-making.
- Facilitates error analysis and model debugging.
- Supports regulatory and ethical requirements for trustworthy artificial intelligence.

Integrating explainability into medical image classification improves the practical usefulness of deep learning systems by enabling healthcare professionals to verify that predictions are based on clinically meaningful image features rather than irrelevant background information.

## 2.6 Gradient-weighted Class Activation Mapping (Grad-CAM)

Gradient-weighted Class Activation Mapping (Grad-CAM) is one of the most widely used explainability techniques for convolutional neural networks. Grad-CAM generates a visual heatmap highlighting the regions of an input image that contribute most strongly to a specific class prediction.
Unlike traditional visualization methods, Grad-CAM utilizes the gradients flowing into the final convolutional layer to estimate the importance of different spatial locations. The resulting heatmap is then superimposed on the original image, allowing users to observe where the model focused when making its prediction.

In dermatological image analysis, Grad-CAM helps verify whether the model is attending to clinically relevant lesion regions instead of unrelated background areas. This improves the interpretability of the classification system and increases confidence in its predictions.
In the proposed framework, Grad-CAM is employed after classification to visualize the regions responsible for the predicted skin condition and severity level. These visual explanations assist both researchers and healthcare professionals in understanding the behavior of the trained deep learning models.

## 2.7 Related Work

Recent advances in deep learning have significantly improved the automatic classification of skin diseases from clinical and dermoscopic images. Researchers have explored various convolutional neural network (CNN) architectures, transfer learning techniques, and Explainable Artificial Intelligence (XAI) methods to improve diagnostic accuracy while increasing the transparency of deep learning models.
Liu et al.[1] developed a deep learning system for the differential diagnosis of skin diseases using clinical images. Their study demonstrated that deep learning models can achieve dermatologist-level performance in recognizing multiple skin diseases, highlighting the potential of artificial intelligence as a clinical decision support tool. However, the proposed system primarily focused on disease classification and did not estimate the severity of skin conditions.
Srinivasu et al.[3] proposed a skin disease classification framework using MobileNetV2 combined with Long Short-Term Memory (LSTM) networks. Their transfer learning approach achieved promising classification performance while reducing computational complexity. Nevertheless, the proposed framework concentrated solely on disease identification without providing information regarding severity levels.
Goceri[2] reviewed the application of deep learning and mobile technologies in dermatology, emphasizing the growing importance of artificial intelligence in supporting dermatologists through automated diagnosis and remote healthcare services. The study highlighted the potential of mobile-based skin disease detection systems while identifying the need for more reliable and interpretable AI models.
Barbieri et al.[4] proposed an artificial intelligence framework for leprosy diagnosis using skin lesion images together with demographic and clinical information. Their work demonstrated the effectiveness of combining image analysis with additional patient information to improve diagnostic performance. However, the framework was designed specifically for leprosy and did not address generalized skin condition severity estimation.
Inthiyaz et al.[5] developed a deep learning-based skin disease detection framework using convolutional neural networks. Their research achieved encouraging classification performance for multiple skin diseases and demonstrated the effectiveness of transfer learning in dermatological image analysis. Similar to many previous studies, the research primarily concentrated on disease classification.
Surasinghe et al.[6] introduced an EfficientNet-B3-based framework integrated with Grad-CAM for detecting Neglected Tropical Skin Diseases. Their model achieved approximately 91.53% classification accuracy while providing visual explanations that highlighted clinically relevant lesion regions. The incorporation of Grad-CAM improved model transparency; however, the study focused only on disease classifica-

tion rather than severity estimation.
Badhon et al.[7] proposed an explainable skin disease classification framework using transfer learning and Grad-CAM visualization. The study evaluated VGG16 and VGG19 models with extensive data augmentation and achieved high classification accuracy on a balanced dataset. The Grad-CAM visualizations enhanced the interpretability of the model predictions. Despite these contributions, the framework did not estimate the severity levels of skin conditions.

| **Study** | **Transfer Learning** | **Explainable AI (Grad-CAM/XAI)** | **Severity Estimation** | **Multiple Models Evaluated** |
|---|---|---|---|---|
| Liu et al. (2020) [1] | ✓ | × | × | × |
| Srinivasu et al. (2021) [3] | ✓ | × | × | × |
| Surasinghe et al. (2023) [6] | ✓ | ✓ | × | × |
| Badhon et al. (2024) [7] | ✓ | ✓ | × | ✓ |
| **Proposed Framework** | ✓ | ✓ | ✓ | ✓ |

Table 2.1: Comparative Analysis of Related Studies and the Proposed Framework

Table 2.1 summarizes the key characteristics of the reviewed studies and compares them with the proposed framework. Most existing studies employ transfer learning for skin disease classification, and some integrate Explainable Artificial Intelligence techniques such as Grad-CAM to improve model interpretability. However, the majority of these approaches focus only on disease classification and do not estimate the severity of skin conditions. Furthermore, only a limited number of studies compare multiple transfer learning architectures under identical experimental settings. The proposed framework addresses these limitations by integrating transfer learning, severity estimation, Grad-CAM-based explainability, and comparative evaluation of multiple deep learning models within a unified system.
Overall, existing studies demonstrate that transfer learning significantly improves skin disease classification performance and that explainability techniques such as Grad-CAM enhance model transparency. However, most previous research has concentrated on identifying disease categories rather than simultaneously estimating disease severity within an explainable deep learning framework.

## 2.8 Research Gap

A review of the existing literature reveals several important research gaps.
First, most existing deep learning studies focus primarily on identifying the type of skin disease. Although these approaches achieve high classification accuracy, they generally provide only the predicted disease category and do not estimate the severity of the detected condition. Consequently, users and healthcare professionals receive limited information regarding the progression or current stage of the skin condition.
Second, many existing explainable AI studies integrate Grad-CAM or similar visualization techniques to improve model interpretability. While these methods successfully highlight the image regions responsible for model predictions, they are generally applied only to disease classification tasks and not to severity estimation.
Third, relatively few studies evaluate multiple transfer learning architectures under identical experimental settings. As a result, selecting the most suitable model for balancing classification performance and computational efficiency remains challenging.

Finally, existing research rarely considers the practical application of severity estimation for long-term treatment monitoring. A system capable of predicting both the skin condition and its severity could assist users and healthcare professionals in observing whether a condition is improving, remaining stable, or worsening over time.

To address these limitations, this research proposes an explainable transfer learning-based framework that simultaneously classifies multiple skin conditions and estimates their corresponding severity or condition levels. Three state-of-the-art transfer learning models—EfficientNetB0, MobileNetV2, and ResNet50—are comparatively evaluated using a common experimental framework. Furthermore, Grad-CAM visualization is integrated to improve the transparency of model predictions. The proposed framework aims to provide more informative decision support by combining classification, severity estimation, and explainability within a unified system.

## 2.9 Chapter Summary

This chapter presented the theoretical background necessary for understanding the proposed research. It discussed skin conditions and severity estimation, deep learning for medical image classification, transfer learning, Explainable Artificial Intelligence, and Grad-CAM visualization. These concepts provide the foundation for the proposed explainable transfer learning framework developed in this study. The related work and research gap will further establish the motivation for the proposed methodology. The next chapter describes the research methodology, including dataset preparation, model development, training procedures, explainability, and performance evaluation.

# Chapter 3

# Research Methodology

## 3.1 Introduction

This chapter describes the methodology adopted to develop the proposed explainable deep learning framework for skin condition classification and severity estimation. The methodology consists of dataset collection, preprocessing, transfer learning model development, training, prediction, explainability, and performance evaluation. Three pre-trained convolutional neural network architectures, namely EfficientNetB0, MobileNetV2, and ResNet50, were implemented and comparatively evaluated. Grad-CAM was incorporated to provide visual explanations of the model's predictions, improving transparency and interpretability.

## 3.2 Research Framework

The proposed framework follows a sequential workflow beginning with dataset collection and preprocessing. The dataset is then divided into training, validation, and testing subsets before applying data augmentation techniques. Three transfer learning models are trained independently using the prepared dataset. After training, each model predicts the probabilities of all skin condition–severity labels. Finally, Grad-CAM visualization is generated to highlight the image regions contributing to the model's predictions, followed by quantitative performance evaluation and comparative analysis.

## 3.3 Dataset Collection

The dataset used in this study consists of 8,275 images grouped into four categories: Acne [8], Burn [10], Eczema [8], and Normal [9] skin. The Acne, Burn, and Eczema categories each contain three severity levels: Mild, Moderate, and Severe. The Normal skin category contains three skin types: Dry, Normal, and Oily, resulting in a total of 12 labels. All images were resized to a uniform resolution of $224 \times 224$ pixels with three color channels (RGB) to maintain consistency across the dataset. To support systematic data management and model development, the dataset was organized into structured folders, and metadata containing image paths, labels, category information, and corresponding severity levels or skin types was generated




Figure 3.1: Representative Images of Skin Conditions and Their Corresponding Severity Levels in the Dataset

automatically. This metadata was used to map the 12 labels to unique numerical indices and facilitate consistent data loading, preprocessing, training, and evaluation.

## 3.4 Data Preprocessing

Data preprocessing was performed to ensure consistency in image dimensions, color channels, and label representation before model training. All images were resized to $224 \times 224$ pixels while preserving three color channels (RGB). Metadata was generated automatically to record image paths, label names, and their corresponding numerical indices, supporting consistent data loading and label mapping throughout the training and evaluation processes.
The dataset contains 12 labels covering three severity levels each for Acne, Burn, and Eczema, along with three Normal skin types: Dry, Normal, and Oily. Each label was assigned a unique numerical index to support the model's output representation. Image validation was also performed to identify and exclude invalid image files, helping maintain dataset integrity.

## 3.5 Dataset Splitting

The processed dataset was divided into three mutually exclusive subsets using different training proportions, ranging from 60% to 90%. For each configuration, the remaining data was divided equally between the validation and test sets. The evaluated configurations were as follows:

- Training Set: 60%, Validation Set: 20%, Test Set: 20%
- Training Set: 70%, Validation Set: 15%, Test Set: 15%
- Training Set: 80%, Validation Set: 10%, Test Set: 10%
- Training Set: 90%, Validation Set: 5%, Test Set: 5%

The training set was used to optimize the model parameters, while the validation set was used to monitor model performance during training, support model selection, and guide adjustments to the training process. The test set was reserved for evaluating model performance on data not used for training or validation. These configurations were investigated to assess how different training data proportions affected classification performance and to identify a suitable dataset split for the proposed framework. For fair comparison among the three transfer learning models, the same split configuration and corresponding data partitions were used when comparing the models under each experimental setting.

## 3.6 Data Augmentation

To improve model generalization and reduce overfitting, data augmentation techniques were applied to the training dataset. The augmentation process included horizontal flipping only.
This transformation generated diverse training samples while preserving the semantic

characteristics of the skin conditions. Validation and test datasets were not augmented to ensure unbiased evaluation.

## 3.7 Proposed Deep Learning Framework

The proposed framework employs transfer learning to adapt pre-trained convolutional neural network architectures for skin condition classification and severity estimation. Instead of training the networks from scratch, ImageNet-pretrained models are used as feature extractors and adapted to the target dataset. Data augmentation and model-specific preprocessing are applied to the input images before feature extraction. A custom classification head is added to the backbone network. It consists of a Global Average Pooling layer, Batch Normalization, a Dropout layer with a rate of 0.30, a fully connected Dense layer containing 256 units with Swish activation and L2 regularization, a second Dropout layer with a rate of 0.30, and an output layer containing 12 units. The output layer uses sigmoid activation to produce an individual score for each skin-condition.
The models are compiled using the Binary Cross-Entropy loss function and the selected optimizer. The sigmoid activation and Binary Cross-Entropy loss support independent output scores for the labels, provided that the target encoding is consistent with a multilabel learning formulation. These scores allow the model's predictions to be examined across all 12 labels rather than displaying only the highest-scoring label.

## 3.8 Transfer Learning Models

Three pre-trained convolutional neural network architectures were selected for comparison:

- EfficientNetB0
- MobileNetV2
- ResNet50

These models were chosen because they employ different architectural approaches to feature extraction and computational efficiency, allowing their suitability for skin condition classification and severity estimation to be evaluated.
EfficientNetB0 uses a compound scaling strategy to balance network depth, width, and input resolution, making it a suitable candidate for investigating the trade-off between classification performance and computational cost. MobileNetV2 employs inverted residual blocks and linear bottlenecks to reduce computational requirements, making it useful for evaluating inference efficiency. ResNet50 uses residual connections to facilitate the training of deeper networks and learn complex visual features, providing a contrasting approach for skin image analysis.
Each model was initialized using ImageNet pre-trained weights. During the first training stage, the feature extraction layers were frozen while only the newly added classification layers were trained. During the second stage, selected layers of the backbone network were unfrozen and fine-tuned using a smaller learning rate to improve task-specific feature learning.

The performance of all three architectures was compared using identical preprocessing, training, and evaluation procedures.

## 3.9 Model Training

The models were trained using the TensorFlow and Keras deep learning framework. Early stopping and model checkpointing were employed to prevent overfitting and preserve the best-performing model based on validation performance.
The Adam optimizer was used for optimization with adaptive learning rate scheduling. ReduceLROnPlateau automatically decreased the learning rate whenever validation performance stopped improving, allowing the model to converge more effectively.
Training performance was monitored using training and validation accuracy together with training and validation loss.

## 3.10 Prediction and Severity Estimation

After training, each model predicts confidence scores for all skin condition–severity labels simultaneously. The prediction results are organized according to their corresponding skin conditions, allowing users to observe the confidence associated with every severity level.
The predicted confidence scores provide a comprehensive assessment rather than only displaying the highest predicted class. This information may assist in monitoring changes in skin condition severity across multiple observations captured during different stages of treatment.

## 3.11 Explainability Using Grad-CAM

To improve transparency, Gradient-weighted Class Activation Mapping (Grad-CAM) was integrated into the proposed framework. Grad-CAM generates heatmaps highlighting the regions of the input image that contribute most significantly to each predicted label.
Unlike conventional classification systems that provide only prediction scores, Grad-CAM allows users to visualize the evidence supporting the model's decisions. This improves the interpretability of the framework and increases user confidence in automated predictions.

## 3.12 Performance Evaluation

The proposed models were evaluated using the independent test dataset. Performance was assessed using several evaluation metrics, including Accuracy, Precision, Recall, and F1-score.
In addition to numerical evaluation, confusion matrices and classification reports were generated to analyze model performance for individual classes. The three transfer learning architectures were then comparatively analyzed to determine the most suitable model for the proposed skin condition classification framework.

## 3.13 Chapter Summary

This chapter presented the methodology used to develop the proposed explainable transfer learning framework for skin condition classification and severity estimation. The methodology covered dataset preparation, preprocessing, transfer learning model development, training, prediction, Grad-CAM visualization, and performance evaluation. The following chapter presents the experimental results and comparative analysis of the proposed models.

# Chapter 4

# Results and Discussion

## 4.1 Introduction

This chapter presents the experimental results of the proposed explainable transfer learning framework for skin condition classification and severity estimation. Three pre-trained convolutional neural network (CNN) architectures—EfficientNetB0, MobileNetV2, and ResNet50—were evaluated under four dataset split configurations. The models were assessed using Accuracy, Precision, Recall, F1-score, ROC-AUC, and inference time per image. Grad-CAM was also employed to visualize image regions associated with model predictions. The results are presented through quantitative evaluation, comparative analysis, and qualitative examination of model explanations.

## 4.2 Experimental Setup

The experiments used a processed skin condition dataset containing 8,275 images categorized into four main groups: Acne, Burn, Eczema, and Normal skin. These groups were represented by 12 labels, comprising mild, moderate, and severe levels for Acne, Burn, and Eczema, together with dry, normal, and oily skin types.
Four dataset split configurations were considered: 60:20:20, 70:15:15, 80:10:10, and 90:5:5, representing the percentages allocated to training, validation, and testing, respectively. Each of the three CNN architectures was associated with these four configurations, resulting in twelve model configurations.
The model names follow the convention `architecture_train_validation_test`. For example, `resnet50_90_5_5` denotes ResNet50 with 90% of the data allocated to training, 5% to validation, and 5% to testing.
Model performance was assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Inference time per image was measured to compare computational efficiency. Grad-CAM visualizations were generated to examine the image regions associated with model predictions.

## 4.3 Training Performance

The training and validation accuracy and loss curves were examined to assess the learning behavior of the twelve model configurations. Accuracy curves provide information about changes in classification performance during training, whereas

loss curves help illustrate the optimization process. Differences between training and validation behavior can also provide indications of potential overfitting.
Figure 4.1 presents the training and validation accuracy curves, while Figure 4.2 shows the corresponding loss curves.

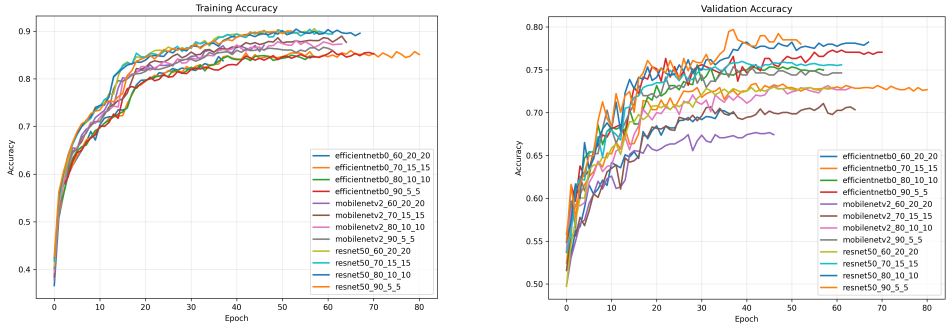



Figure 4.1: Training and Validation Accuracy Curves

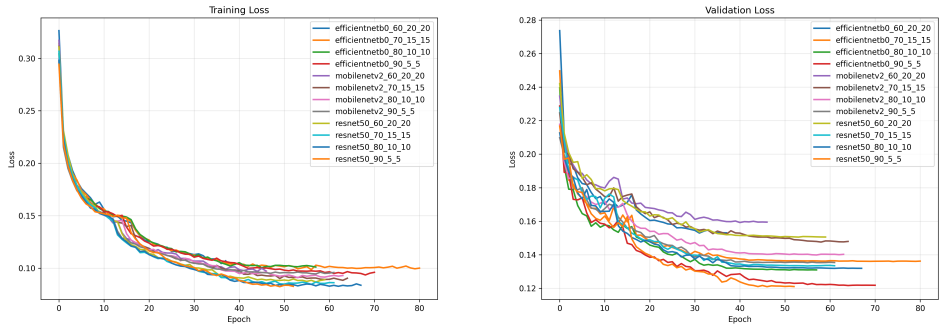



Figure 4.2: Training and Validation Loss Curves

The learning behavior of each model should be interpreted in relation to its architecture and dataset split. A larger training partition provides more examples for learning discriminative features, while the validation partition supports monitoring and model selection. However, training curves alone are insufficient to establish generalization performance. The quantitative evaluation and the validity of the test procedure must also be considered when comparing the models.

## 4.4 Performance Evaluation

All twelve models were compared using Accuracy, Precision, Recall, F1-score, ROC-AUC, and inference time per image. The reported results are presented in Table 4.1. According to the reported results, `ResNet50_90_5_5` achieved the highest values for Accuracy, Precision, Recall, F1-score, and ROC-AUC. Its results were 86.84%, 87.22%, 86.84%, 86.79%, and 99.27%, respectively. Among the remaining configurations, `EfficientNetB0_90_5_5` achieved an Accuracy of 85.02%, while `MobileNetV2_90_5_5` achieved 83.21%.
The lowest reported Accuracy was 67.03%, obtained by `MobileNetV2_60_20_20`. The results therefore show variation in classification performance across both the CNN architectures and the dataset configurations.

| Model | Accuracy (%) | Precision (%) | Recall (%) | F1-score (%) | ROC-AUC (%) | Inference Time (ms/image) |
|---|---|---|---|---|---|---|
| EfficientNetB0_60_20_20 | 69.81 | 69.89 | 69.81 | 69.62 | 96.67 | 26.13 |
| EfficientNetB0_70_15_15 | 73.19 | 73.44 | 73.19 | 72.94 | 97.26 | 24.26 |
| EfficientNetB0_80_10_10 | 74.88 | 74.98 | 74.88 | 74.70 | 97.49 | 26.16 |
| EfficientNetB0_90_5_5 | 85.02 | 85.17 | 85.02 | 84.88 | 98.94 | 23.27 |
| MobileNetV2_60_20_20 | 67.03 | 67.49 | 67.03 | 66.83 | 96.59 | 16.36 |
| MobileNetV2_70_15_15 | 71.62 | 71.88 | 71.62 | 71.46 | 97.33 | 16.15 |
| MobileNetV2_80_10_10 | 73.91 | 74.04 | 73.91 | 73.59 | 97.54 | 16.54 |
| MobileNetV2_90_5_5 | 83.21 | 83.71 | 83.21 | 83.16 | 98.72 | 18.53 |
| ResNet50_60_20_20 | 72.83 | 73.16 | 72.83 | 72.43 | 97.23 | 55.82 |
| ResNet50_70_15_15 | 76.09 | 76.37 | 76.09 | 75.81 | 97.91 | 54.03 |
| ResNet50_80_10_10 | 78.50 | 78.97 | 78.50 | 78.34 | 98.02 | 66.58 |
| ResNet50_90_5_5 | **86.84** | **87.22** | **86.84** | **86.79** | **99.27** | 72.42 |

Table 4.1: Performance Comparison of All Twelve Transfer Learning Models

Inference time also differed across the models. `MobileNetV2_70_15_15` recorded the shortest time at 16.15 ms per image, whereas `ResNet50_90_5_5` recorded the longest at 72.42 ms per image. These measurements indicate that classification performance and inference speed should be considered together when selecting a model for a particular application.

## 4.5 Comparative Analysis

This section compares EfficientNetB0, MobileNetV2, and ResNet50 across the four evaluated dataset split configurations: 60:20:20, 70:15:15, 80:10:10, and 90:5:5. The comparison considers Accuracy, Precision, Recall, F1-score, ROC-AUC and inference time to examine both classification performance and computational efficiency. Figure 4.3 provides a visual summary of these metrics across the evaluated configurations.

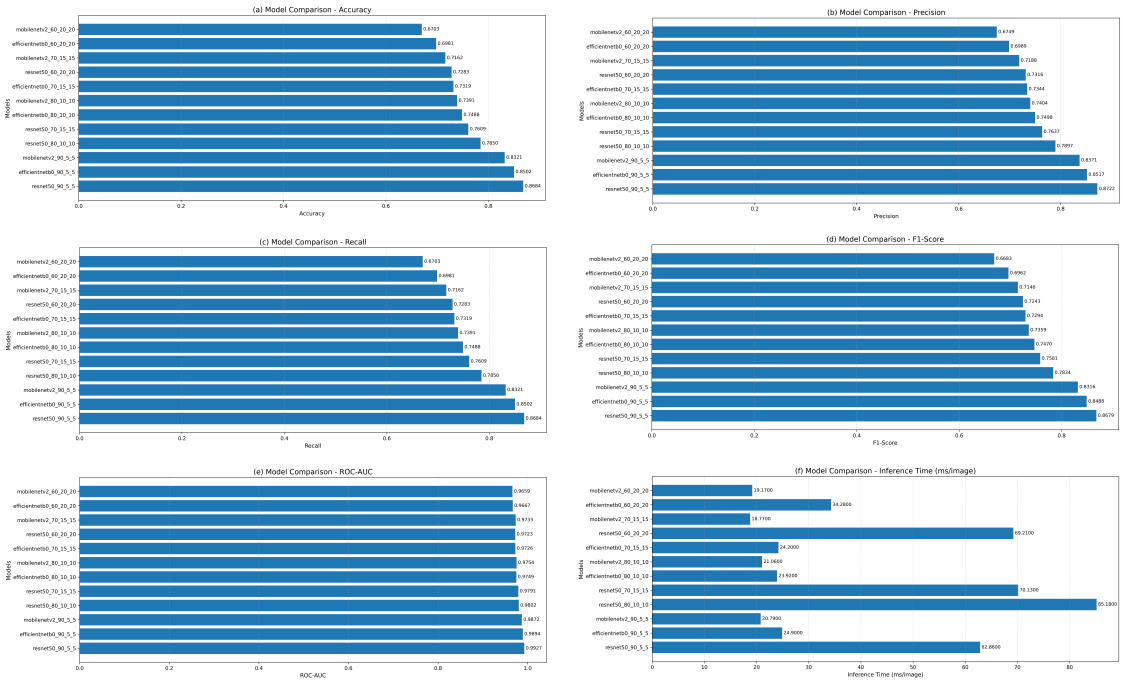



Figure 4.3: Comparison of EfficientNetB0, MobileNetV2, and ResNet50 across four dataset split configurations in terms of (a) Accuracy, (b) Precision, (c) Recall, (d) F1-score, (e) ROC-AUC and (f) inference time.

### 4.5.1 Comparison Across Dataset Split Configurations

The reported Accuracy increased across the four split configurations for each of the three architectures. For EfficientNetB0, Accuracy increased from 69.81% under 60:20:20 to 74.88% under 80:10:10, before reaching 85.02% under 90:5:5. MobileNetV2 followed a similar pattern, increasing from 67.03% to 73.91% across the first three configurations and reaching 83.21% under 90:5:5. ResNet50 increased from 72.83% under 60:20:20 to 78.50% under 80:10:10 and 86.84% under 90:5:5.

This pattern suggests that the configurations with larger training proportions were associated with higher reported Accuracy. One possible explanation is that a larger training set provides more examples from which the model can learn useful image representations. Nevertheless, the observed differences cannot be attributed solely to the amount of training data without ensuring that the evaluation procedure is consistent across configurations.

The 90:5:5 configuration also allocates smaller proportions of the dataset to validation and testing. A smaller validation set may provide less information for model selection, while a smaller test set can lead to less stable performance estimates. Accordingly, the results do not establish that 90:5:5 is universally the best split configuration.

### 4.5.2 Comparison of CNN Architectures

The three architectures exhibited different balances between classification performance and inference time. ResNet50 achieved the highest reported Accuracy under each of the four split configurations. Its stronger reported classification performance was accompanied by longer inference times, particularly for the 90:5:5 configuration.

EfficientNetB0 provided a potential middle ground. Its `90_5_5` configuration achieved 85.02% Accuracy with an inference time of 23.27 ms per image. Although this was lower in Accuracy than the corresponding ResNet50 configuration, its measured inference time was substantially shorter.

MobileNetV2 recorded the shortest inference times among the three architectures. Its fastest configuration, `MobileNetV2_70_15_15`, required 16.15 ms per image. Its reported classification performance was lower than that of the corresponding EfficientNetB0 and ResNet50 configurations, but its speed may make it a useful candidate for applications with limited computational resources.

Overall, the reported results suggest that ResNet50 is a candidate when classification performance is the primary consideration, EfficientNetB0 may provide a compromise between performance and latency, and MobileNetV2 may be suitable where computational efficiency is particularly important. These conclusions should be considered provisional until the evaluation partitions are verified.

## 4.6 Explainability Using Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) was incorporated to provide a qualitative view of the image regions associated with model predictions. It produces heatmaps using gradients associated with a target class, helping visualize which spatial regions contribute to the model's output.

The generated heatmaps can be examined to determine whether the model's attention overlaps with relevant skin regions or appears to be influenced by background

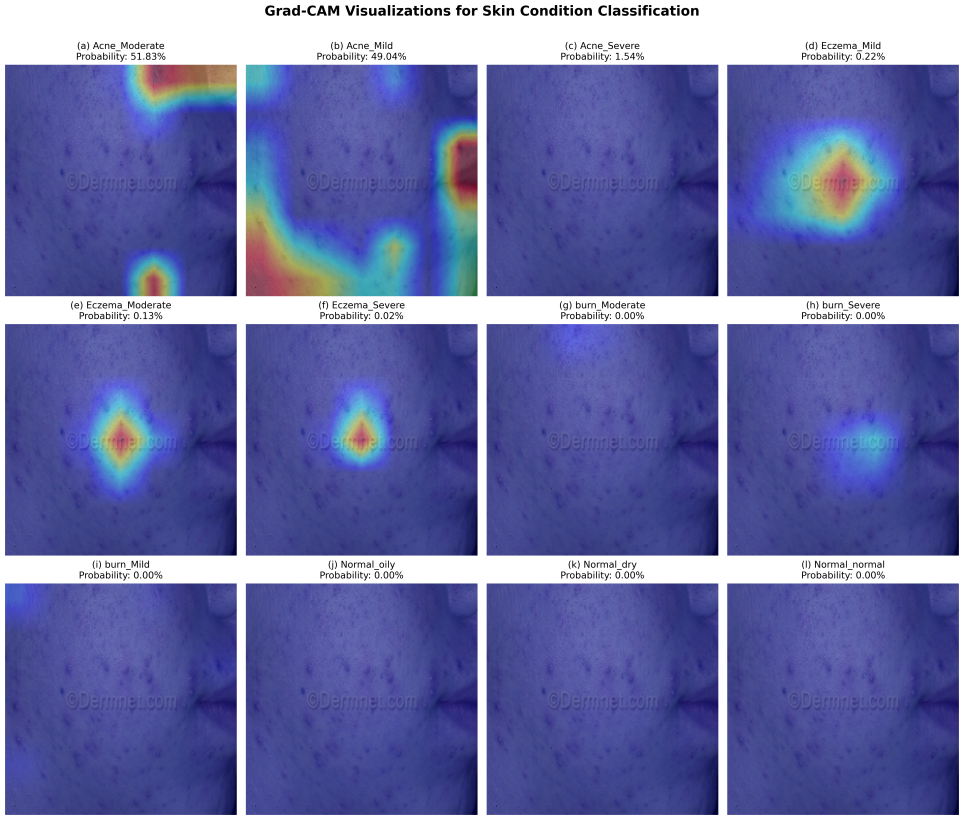



Figure 4.4: Grad-CAM Visualizations for Skin Condition Classification

content. Features such as lesion boundaries, inflammation, and pigmentation should be discussed as observed findings only when the visualizations provide clear support for those interpretations.

Grad-CAM complements the quantitative evaluation by making aspects of model behavior easier to inspect. However, a heatmap does not prove that a model relies on clinically valid features or that its prediction is medically correct. These visualizations should therefore be interpreted as qualitative explanations rather than evidence of diagnostic validity.

## 4.7 Discussion

The results show differences among EfficientNetB0, MobileNetV2, and ResNet50 in classification performance and computational efficiency. `ResNet50_90_5_5` achieved the highest reported Accuracy (86.84%) and ROC-AUC (99.27%), whereas `MobileNetV2_70_15_15` recorded the shortest inference time (16.15 ms per image). These findings highlight the trade-off between classification performance and prediction speed when selecting a model.

Accuracy increased across the evaluated configurations as the training proportion increased. Although additional training images may help models learn more representative features, differences in validation and test set sizes may also affect the reported results. Therefore, the 90:5:5 configuration cannot be considered universally optimal based on these experiments alone.

Grad-CAM provides visual explanations of image regions associated with model

predictions, but these visualizations do not establish clinical reliability. Further evaluation on diverse, independent datasets is needed to assess generalization and determine the framework's suitability for potential healthcare applications.

## 4.8 Chapter Summary

This chapter presented the reported results for twelve transfer learning configurations based on EfficientNetB0, MobileNetV2, and ResNet50. ResNet50_90_5_5 achieved the strongest reported classification metrics, while MobileNetV2_70_15_15 recorded the shortest inference time. The comparison illustrates the importance of balancing predictive performance with computational requirements. Grad-CAM was also used to support qualitative inspection of model predictions.
The principal methodological consideration is to verify that each model was evaluated on the appropriate independent test partition and that the evaluation metrics and inference times were calculated consistently. Resolving this issue is necessary before making definitive claims about the influence of dataset split proportions or selecting a model for practical deployment.

# Chapter 5

# Conclusion and Future Work

## 5.1 Conclusion

This research developed an explainable transfer learning framework for automated skin condition classification and severity estimation. The framework combines deep learning-based image classification with Gradient-weighted Class Activation Mapping (Grad-CAM) to provide predictions alongside visual explanations of the image regions influencing those predictions. The primary objective was to investigate the classification performance, computational efficiency, and interpretability of different transfer learning architectures for skin image analysis.

A complete image preprocessing pipeline was implemented to support dataset preparation, image resizing, metadata generation, dataset partitioning, and data augmentation. Three pre-trained convolutional neural network architectures—EfficientNetB0, MobileNetV2, and ResNet50—were adapted using transfer learning and evaluated across different training, validation, and testing configurations. This experimental setup enabled a comparative investigation of model performance under the selected configurations.

The experimental results demonstrated that EfficientNetB0, MobileNetV2, and ResNet50 achieved promising classification performance across the evaluated configurations. Their comparative evaluation revealed differences in predictive performance and inference time. ResNet50 achieved the strongest classification performance in the reported results, while EfficientNetB0 and MobileNetV2 offered alternatives with lower inference times. These findings highlight the importance of considering both classification performance and computational efficiency when selecting an architecture for a particular application.

In addition to classification, the framework generates confidence scores for the available skin condition and severity labels, allowing users to examine the model's relative predictions rather than relying solely on the highest-ranked class. Grad-CAM visualizations further provide a qualitative view of the image regions associated with a prediction. Together, these features improve the transparency of the system's outputs, although neither confidence scores nor visual explanations independently establish the clinical correctness of a prediction.

Overall, this research demonstrates the potential of combining transfer learning with explainable artificial intelligence for automated skin image analysis. The proposed framework may serve as a foundation for computer-aided assessment and further research into skin condition classification and severity estimation. However, broader

validation, particularly using diverse real-world images and expert-annotated clinical data, is necessary before the framework can be considered for practical healthcare applications.

## 5.2 Contributions of the Research

The main contributions of this research are summarized as follows:

- An explainable deep learning framework was developed for skin condition classification and severity-related label prediction.
- An image preprocessing pipeline was implemented to support image resizing, metadata generation, dataset preparation, partitioning, and augmentation.
- Three transfer learning architectures—EfficientNetB0, MobileNetV2, and ResNet50—were evaluated to compare their classification performance and computational efficiency.
- Multiple train-validation-test configurations were investigated to examine model performance under different data allocation strategies.
- Confidence scores for the available condition and severity labels were incorporated to provide more information about the model's predictions than a single predicted label alone.
- Grad-CAM visualization was integrated to provide qualitative explanations of the image regions associated with model predictions.
- Classification metrics and inference time were considered together to identify trade-offs between predictive performance and computational cost.

## 5.3 Limitations

Despite the promising results, this study has several limitations. First, the dataset covers a limited set of skin conditions and condition-related labels. Consequently, the findings should not be interpreted as evidence that the framework can recognize the full range of dermatological diseases or severity levels encountered in clinical practice.

Second, the performance of image-based classification systems can be affected by differences in skin tone, lighting, camera characteristics, image resolution, viewing angle, and image quality. The extent to which the proposed framework generalizes to these variations requires further investigation using more diverse and representative datasets.

Third, the predicted condition and severity labels depend on the quality and consistency of the dataset annotations. The model's ability to distinguish between severity levels therefore requires evaluation against reliable reference labels, ideally established or reviewed by qualified dermatologists. In addition, confidence scores and Grad-CAM visualizations do not, by themselves, guarantee calibrated uncertainty or clinically valid explanations.

Finally, the experimental results depend on the dataset partitioning and evaluation protocol used for each model. Fair comparisons between different train-validation-test configurations require each model to be evaluated using the appropriate held-out test data under a consistent protocol. The reported inference times may also vary with the hardware, input dimensions, batch size, and measurement procedure.
Accordingly, the proposed framework should be regarded as a research prototype for computer-aided skin image analysis rather than a substitute for professional medical diagnosis. Further technical evaluation and clinical validation are necessary before practical deployment.

## 5.4 Future Work

Several directions can be explored to improve the proposed framework and assess its suitability for real-world applications.

### 5.4.1 Dataset Expansion and Generalization

Future research should expand the dataset to include a broader range of dermatological conditions, severity levels, skin tones, age groups, and imaging environments. Collecting images from multiple healthcare institutions and imaging devices would help evaluate the model's ability to generalize beyond the conditions represented in the current dataset. Expert-annotated data and clearly defined severity criteria would also improve the reliability of model evaluation.

### 5.4.2 Model Architecture and Performance Optimization

Further studies may investigate alternative architectures, including EfficientNetV2, ConvNeXt, Vision Transformers (ViTs), and hybrid convolutional neural network–transformer models. Ensemble learning could also be explored to determine whether combining predictions from multiple architectures improves classification performance. Any performance gains should be assessed alongside inference time, memory requirements, and computational cost to determine the suitability of these approaches for different deployment environments.

### 5.4.3 Enhanced Explainability and Uncertainty Estimation

The explainability component could be extended by evaluating additional techniques, such as Score-CAM, Eigen-CAM, LIME, and SHAP. A systematic comparison of these methods could help determine whether their explanations are consistent, informative, and relevant to the visual characteristics of the input images. Future work should also investigate confidence calibration and uncertainty estimation so that the system can better identify predictions for which it has limited confidence. These measures may help support safer interpretation of model outputs, although they would not replace clinical validation.

### 5.4.4 Mobile and Web-Based Deployment

Another direction is to develop a mobile or web-based application that integrates the trained model with an accessible user interface. The application could allow users to upload or capture skin images and view predicted labels, associated confidence scores, and Grad-CAM visualizations. MobileNetV2 may be a suitable candidate for resource-constrained devices because of its comparatively short inference time in the reported experiments. However, deployment decisions should be based on additional testing under realistic device and network conditions.
Such an application should communicate its limitations clearly and present its outputs as preliminary, model-generated information rather than a confirmed medical diagnosis. Appropriate privacy protections and safeguards for handling user-submitted images would also be necessary.

### 5.4.5 Clinical Validation and Responsible Deployment

Before the framework is considered for clinical use, it should be evaluated using larger, diverse, real-world datasets collected in collaboration with hospitals and dermatology clinics. Prospective studies and independent external validation could help establish how well the system performs on data from sources not represented during training. Evaluation should include class-specific performance, errors across relevant demographic groups, calibration, and the reliability of severity-related predictions. Collaboration with dermatologists would be essential for defining clinically meaningful evaluation criteria and determining how the system could complement existing assessment procedures.

## 5.5 Final Remarks

This research investigated the use of transfer learning and explainable artificial intelligence for skin condition classification and severity-related label prediction. By combining established convolutional neural network architectures with confidence-based outputs and Grad-CAM visualizations, the proposed framework provides both predictive results and a means of inspecting the image regions associated with those results. The comparative evaluation also highlights the importance of balancing classification performance with computational efficiency. Although the findings are encouraging, further validation is required to establish generalization, explanation reliability, and clinical relevance. With these improvements, the framework could contribute to the development of responsible computer-aided tools for dermatological image analysis.

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